ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Coding for DOS

I. Coding for DOS of Pristine Graphene

II. DOS Results based on Smearing scheme and Tetrahedron Method

III. Details of Coursework

I. Coding for DOS of Pristine Graphene

Parameterization and Setup

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Reciprocal Space Transformation and Neighbor Search

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Input Parsing and *k* mesh Generation

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Hamiltonian Construction and Diagonalization

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Multi-Method DOS Calculations

- ♦ Lorentzian Smearing
- ♦ Gaussian Smearing
- ♦ 2D Tetrahedron/Triangle Method

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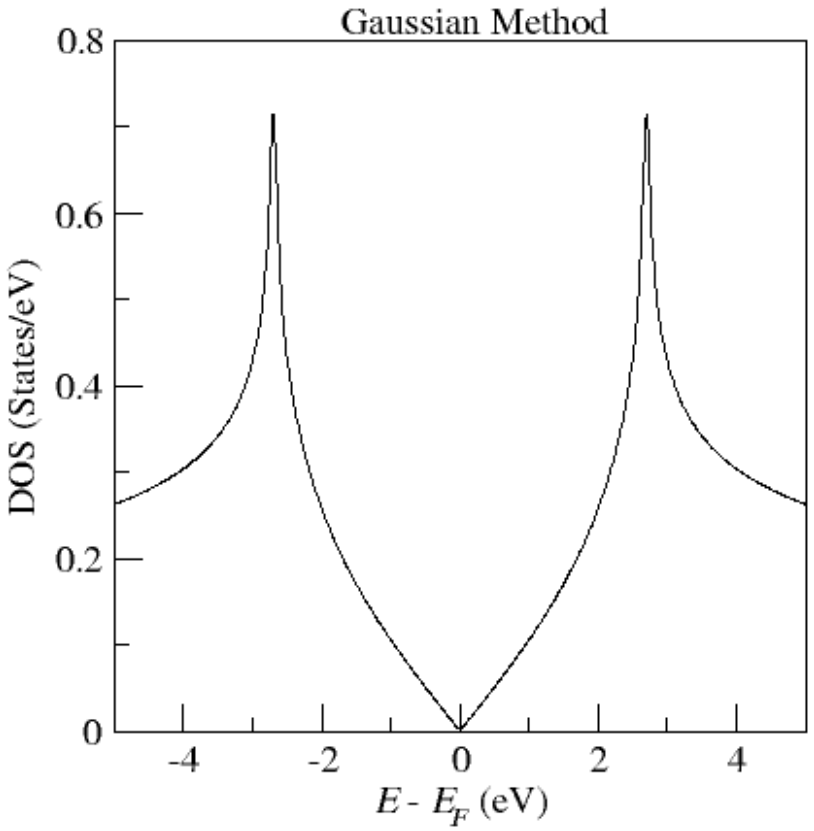
Deallocation and Termination

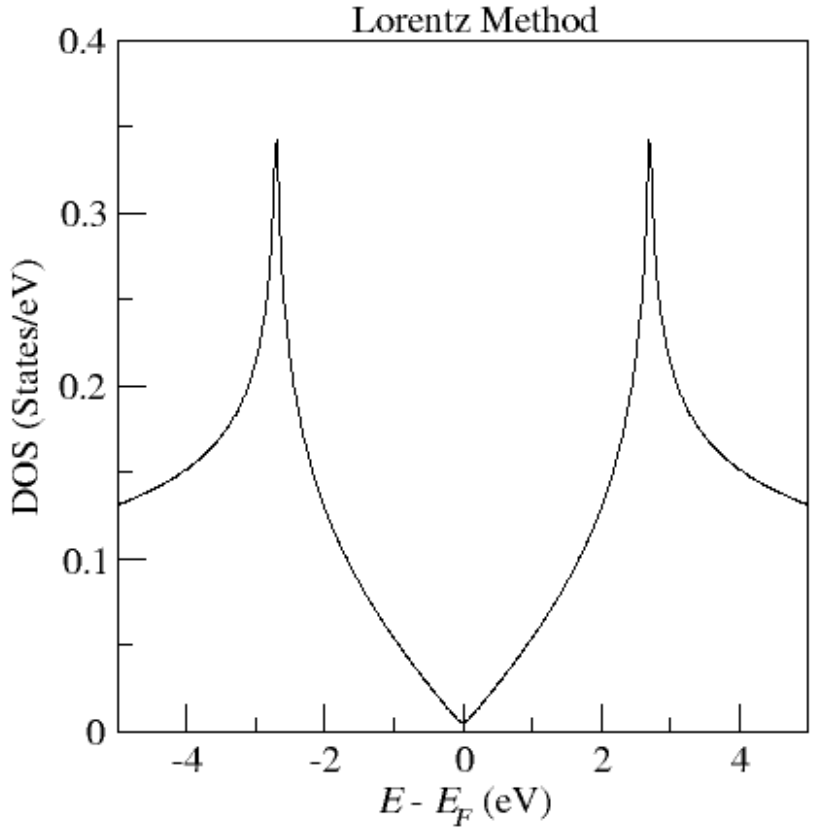
I. Coding for DOS of Pristine Graphene

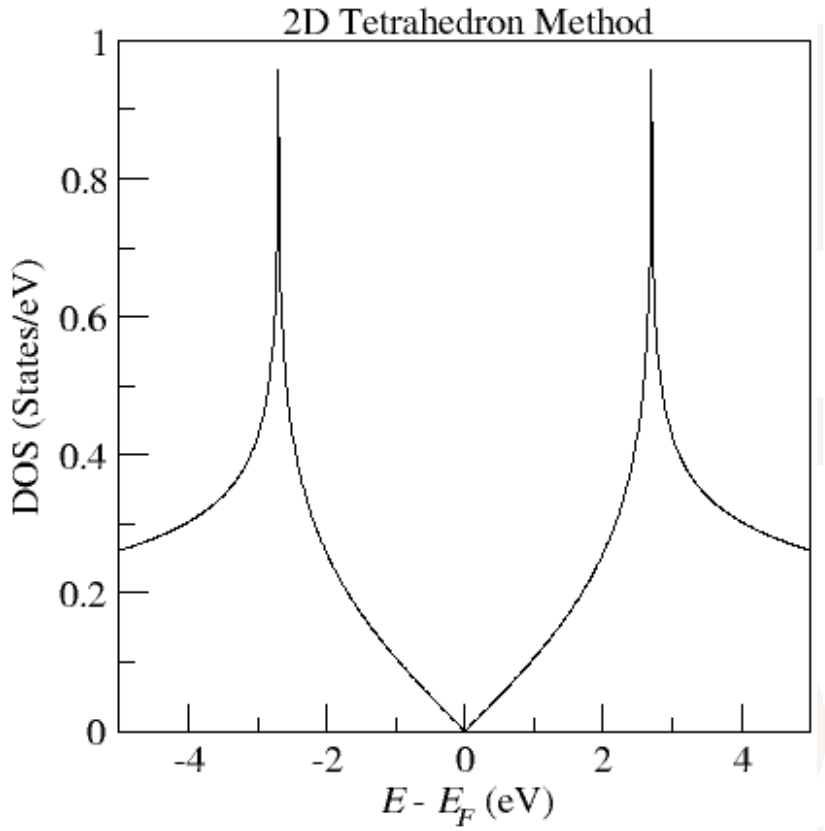
Graphene.vasp File

2.46
Lattice Constant
1.0 0.0 0.0
-0.5 0.8660254037844386 0.0
0.0 0.0 1.0
C
2
Direct
0.6666666666 0.3333333333 0.0
0.3333333333 0.6666666666 0.0
onsite: 0.0 0.0
k point mesh for DOS: 500 500
Energy range for DOS: -5.0 5.0
Energy point for DOS: 1000
Broadening value for DOS: 0.05

II. DOS Results based on Smearing scheme and Tetrahedron Method



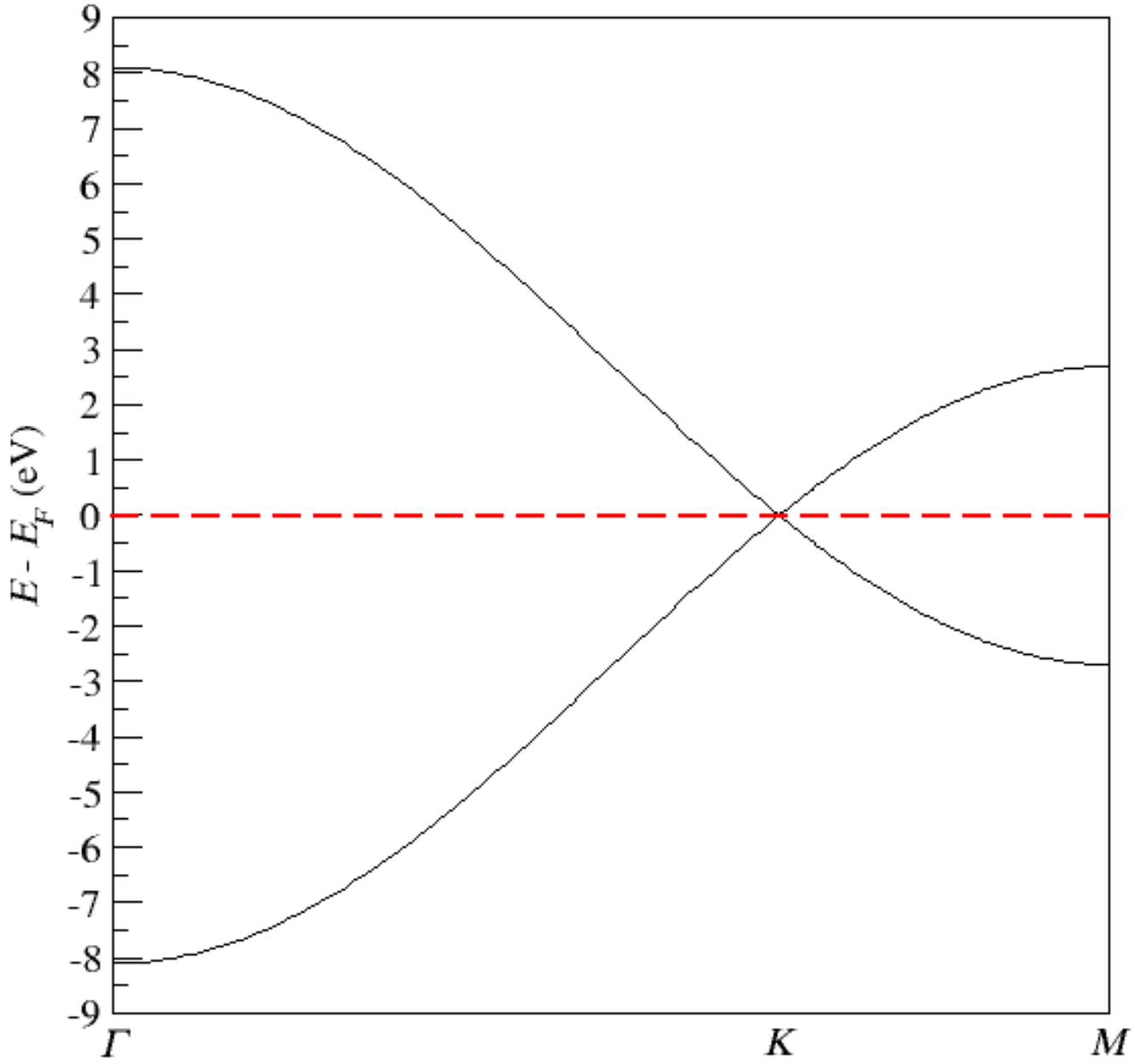




Gaussian Broadening Function: $\delta_{Gaussian}(x,\sigma) = \frac{1}{\sigma\sqrt{\pi}} e^{\left(-\frac{x^2}{\sigma^2}\right)}$

Lorentzian Broadening Function: $\delta_{Lorentz}(x,\Gamma) = \frac{1}{\pi}\frac{\Gamma}{x^2+\Gamma^2}$

II. DOS Results based on Smearing scheme and Tetrahedron Method



III. Details of Coursework

Please, finish the following two coding tasks as your coursework

Coding Task One: Band Structure of Graphene

1. Implement a program to compute the band structure of both pristine and doped graphene
2. Plot the band structures of pristine graphene and doped graphene
3. Interpret the results, *e.g.* the Fermi energy level, the band gap of doped graphene and any other notable features
4. Include the source code behind the corresponding band structure plots

Notes:
1. The program can be written in any of the following languages: Fortran, C, C++, Matlab, Python, or Julia
2. The coursework must be submitted in PDF format
3. There is no requirement on page count or word limit, as long as the above four tasks are clearly addressed

Submission Deadline: End of Week 7 of the semester

III. Details of Coursework

Coding Task Two: Density of States of Pristine Graphene

1. Write a program to compute the density of states (DOS) of pristine graphene using three different methods:
♦ Gaussian smearing scheme
♦ Lorentz smearing scheme
♦ Tetrahedron method
2. Plot the DOS of pristine graphene obtained from each of the three methods
3. Interpret the results, *e.g.* The effect of the broadening parameter, the variation of DOS with energy, Physical properties influenced by the DOS (*e.g.*, conductivity, carrier concentration, etc.)
4. Include the source code behind the corresponding DOS plots

Notes:
1. The program can be written in any of the following languages: Fortran, C, C++, Matlab, Python, or Julia
2. The coursework must be submitted in PDF format
3. There is no requirement on page count or word limit, as long as the above four tasks are clearly addressed

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That's all for this session

Thank you for your attention and great participation